

Router Documentation
Biosensing Platform & ML Classification System
Piezo Pico-to-Femtotechnology Sensors Inc.
Version 1.0 — August 2026

Table of Contents

1. Introduction
2. Router Overview
3. Router Architecture
4. Router–Model Integration
5. Swagger Endpoint Summary
6. Folder Structure Reference
7. Student Workflow Integration
8. Deployment Notes
9. Architecture Diagram
10. Contact and Copyright Information
11. Conclusion

1. Introduction

This Router Documentation describes the backend routing architecture of the **Biosensing Platform & ML Classification System**, developed by **Piezo Pico-to-Femtotechnology Sensors Inc.** It explains how FastAPI routers connect biosensing simulation, machine-learning classification, virus processing, and educational workflows.

This document references two external files:

- **Swagger.pdf** — Full API endpoint specification
- **Folder Structure of Selected ML Models.pdf** — ML directory layout

2. Router Overview

The FastAPI backend uses modular routers to maintain clarity and separation of concerns. Each router handles a specific functional domain.

Router File	Description
<code>simulate_router.py</code>	Generates biosensing features
<code>classify_v2_router.py</code>	Runs V2 (XGBoost) classification
<code>classify_v6_router.py</code>	Runs V6 (Random Forest) classification
<code>classify_spectral_router.py</code>	Runs spectral-signature classification
<code>docs_router.py</code>	Serves documentation references

Additional routers (authentication, payments, courses, virus, training, sensor drift) are defined in **Swagger.pdf**.

3. Router Architecture

3.1 `simulate_router.py`

Endpoint: GET /simulate

Generates biosensing features including:

- Base frequency
- Measured frequency
- Frequency shift
- Derived biosensing features

Used in **Step 1** of the student laboratory workflow.

3.2 `classify_v2_router.py`

Endpoint: GET /classify/v2

Runs the V2 model trained on **12,000 samples**.

Students analyze:

- Noise sensitivity
- Early-stage ML behavior
- Partial virus identification

3.3 classify_v6_router.py

Endpoint: GET /classify/v6

Runs the V6 model trained on **120,000 samples**.

Students analyze:

- High-accuracy predictions
- Full virus identification
- Probability vs mass
- Probability vs virus ID

3.4 classify_spectral_router.py

Endpoint: GET /classify/spectral

Uses FFT-based spectral signatures to classify viruses with unique resonance patterns.

3.5 docs_router.py

Provides:

- Links to Swagger UI
- Links to OpenAPI JSON
- Links to ML documentation
- Links to device documentation
- Links to virus tables

4. Router–Model Integration

Routers load ML models from:

- ml/models/classify/v2/sim_model_v2_100.pkl
- ml/models/classify/v6/sim_model_v6_100.pkl
- ml/models/classify/spectral/spectral_model.pkl

Routers use utilities from:

- ml/utls/feature_extraction.py
- ml/utls/noise_simulation.py
- ml/utls/mass_estimation.py
- ml/utls/preprocessing.py

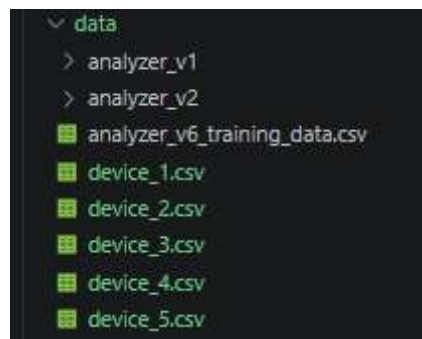
Routers support evaluation via:

- ml/evaluation/compare_v2_v6.py
- ml/evaluation/confusion_matrices/
- ml/evaluation/metrics.py

Routers access virus data from:

- data/virus_table.csv
- data/virus_metadata.json
- data/virus_multiclass.csv
- data/virus_master.csv

Other data tables include:



5. Swagger Endpoint Summary

(Source: Active Swagger Tab — AI Biosensing API — turnobrowsertab1747043343)

Authentication:

- **POST** /auth/register — Register
- **POST** /auth/login — Login
- **GET** /auth/validate-key — Validate Key
- **GET** /auth/users — List Users

Payments:

- **POST** /payments/consulting/fixed
- **POST** /payments/consulting/custom
- **POST** /payments/course
- **POST** /payments/webhook

Course Modules:

- **POST** /course-modules/
- **GET** /course-modules/course/{course_id}
- **PUT** /course-modules/{module_id}
- **DELETE** /course-modules/{module_id}

Course Content:

- **POST** /course-content/
- **GET** /course-content/{course_id}
- **DELETE** /course-content/{id}

Enrollment:

- **POST** /enrollment/{course_id}
- **GET** /enrollment/status/{course_id}
- **GET** /enrollment/my-courses
- **GET** /enrollment/all

Consultations:

- **GET** /consultations/
- **GET** /consultations/{consultation_id}
- **PATCH** /consultations/{consultation_id}/status
- **GET** /consultations/overview

Students & Teams:

- **GET** /students/
- **GET** /teams/

Virus Processing:

- **POST** /virus/ — Create Virus
- **GET** /virus/list — List Viruses
- **GET** /virus/count — Count Viruses
- **POST** /virus/virus/dashboard_data
- **POST** /virus/virus/probabilities/v7

Model Versioning:

- **GET** /versioning/v7

ML Training:

- **GET** /dashboard/ml/train/v2
- **GET** /dashboard/ml/train/v6

Sensor Drift:

- **GET** /sensor/live_init
- **GET** /sensor/live_tick

6. Folder Structure Reference

(Source: Folder Structure of Selected ML Models.pdf)

```
app/  
  main.py  
  routers/  
  ml/  
    models/  
    training/  
    utils/  
    evaluation/  
  data/  
  docs/  
  static/  
  dashboards/
```

7. Student Workflow Integration

Routers support the full student workflow:

- **Simulation** — /simulate
- **V2 Classification** — /classify/v2

- **V6 Classification** — /classify/v6
- **Comparison** — Dashboard visualizations
- **Lab Report** — Students summarize results

8. Deployment Notes

Start API

uvicorn app.main:app --reload

Swagger UI

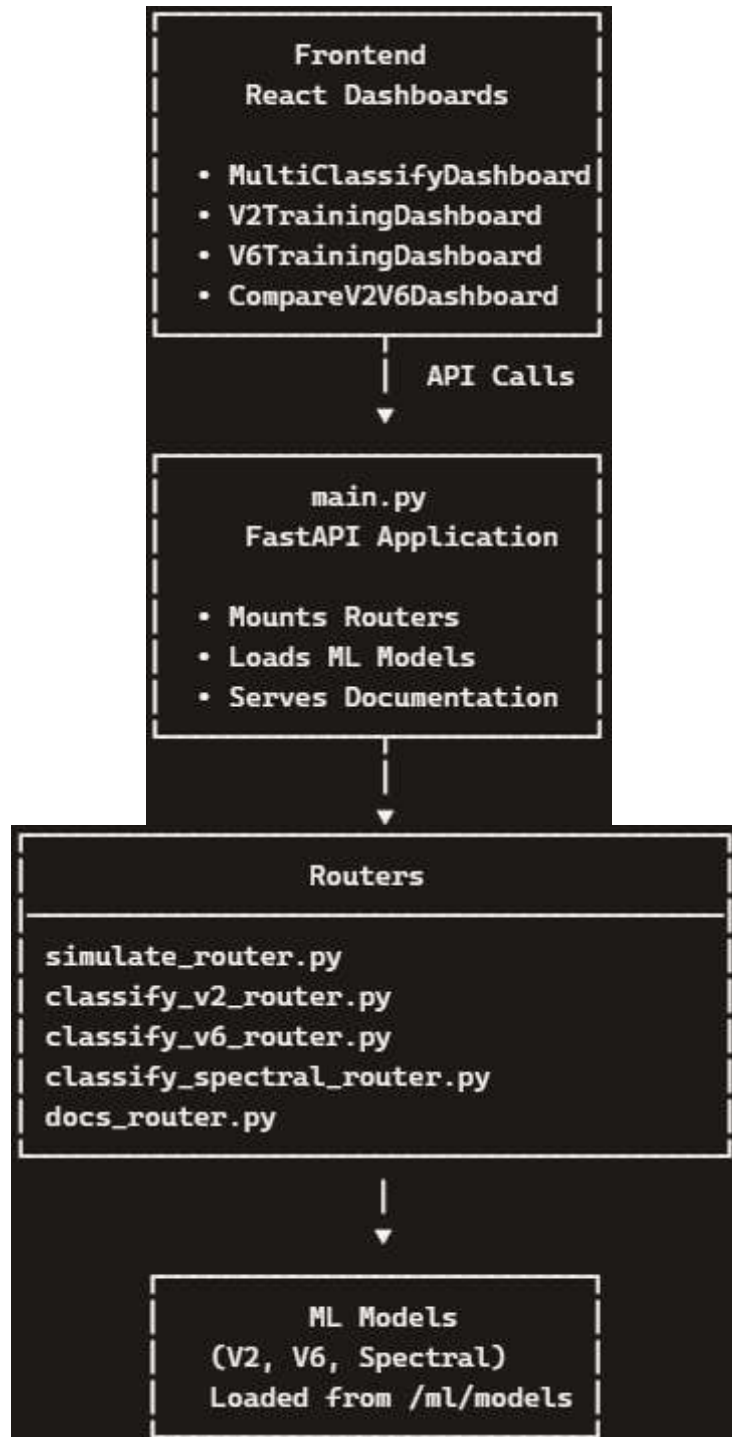
<http://127.0.0.1:8000/docs>

Frontend Integration

Dashboards call: **npm run dev** to run the API from frontend to backend server enabling calling backend from frontend.

- /Simulate
- /Classify/v2
- /Classify/v6
- /Classify/spectral

9. Architecture Diagram



10. Publication & Contact Information

Publisher:

Piezo Pico-to-Femtotechnology Sensors Inc. Edmonton, Alberta, Canada

Contact:

- **Email:** selemaniseif12@yahoo.com or selemaniseif1974@gmail.com
- **Website:** Piezo Pico to Femtotechnology Sensors Inc.
- **Support:** selemaniseif12@yahoo.com
- **Phone:** 780 6044032

Copyright

© 2026 *Piezo Pico-to-Femtotechnology Sensors Inc. All rights reserved.*

11. Conclusion

This Router Documentation completes the backend technical reference for the Biosensing Platform & ML Classification System. It integrates:

- Router architecture
- Swagger endpoints
- ML folder structure
- Student workflow
- Deployment instructions
- System architecture diagram